SCADA-to-CMMS Dispatch Runbook for XG-440 Protective Trips

Model: VFIP-RB-SCADA-CMMS-TRIPS | Category: Procedures

Description: Dispatch runbook mapping specific SCADA alarm codes and telemetry patterns to CMMS corrective work order templates for inverter protective-trip events (including TRIP-31 and TRIP-44). Includes decision trees: immediate actions for safety-critical alarms, threshold-based triage rules, templated work order fields, and escalation contacts. Contains vendor firmware notes, change ticket summaries, O&M incident report, and settlement guidance for ERCOT reconciliation.

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1. Executive Summary

This runbook provides operational procedures to map SCADA protective-trip alarms from XG-440 inverters to CMMS corrective work orders. It is designed for dispatch operators, O&M technicians, and automation engineers integrating SCADA alarm streams with CMMS APIs. The document consolidates vendor firmware release notes for versions v3.14.2 through v3.15.0, vendor change ticket summaries for rollout and hotfix windows, a detailed O&M incident postmortem related to a firmware incident, triage decision trees (including threshold logic for counts per hour and concurrent alarms across the site), templated CMMS fields (priority, labor code, parts SKU), escalation contacts, and settlement guidance for ERCOT imbalance charges.

Key alarm families covered: TRIP-31 (grid ride-through protective trip), TRIP-44 (internal hardware protection trip), and correlated telemetry patterns such as repeated inverter availability loss, repeated restart attempts, and concurrent alarms across multiple inverters.

2. Vendor Firmware Release Notes (v3.14.2 → v3.15.0)

2.1 Release v3.14.2 (Issued: 2025-11-18)

Summary: Stability release addressing protective-trip telemetry noise and improving trip reporting fidelity. Revisions relevant to protective trip handling:

Change	Description	Impact
Trip code mapping	Standardized internal code mapping to SCADA alarm field scada_alarm_code for TRIP-31 and TRIP-44.	Enables consistent automated ticketing based on scada_alarm_code.
Debounce timers	Introduced configurable debounce window parameter *trip_debounce_ms* (default 300ms).	Reduces spurious trip events derived from short transients.
Telemetry tag added	Added field *availability_minutes_lost* to local event logs, exported to SCADA every 10s.	Facilitates settlement reconciliation.

Known Issues

- Under rare conditions high restart attempt count reported while inverter remains unavailable (workaround: force-stop restart loop at controller).
- Change affects alarm rate; operators should adjust triage thresholds accordingly during rollout.

2.2 Release v3.15.0 (Issued: 2025-12-06)

Summary: Hotfix addressing an edge-case where protective trips were incorrectly elevated to hardware fault severity during rapid grid events. Relevant modifications:

Change	Description	Notes
TRIP-44 logic	Refined criteria to differentiate transient internal over-temperature from hardware failure (added sustained temp > 85°C for 30s).	Reduces false-positive hardware replacements.
Event correlation	Added multi-channel correlation flag *trip_correlated* when >1 inverter in site reports protective trip within 120s.	Useful for site-level triage rules.
Logging	Extended event_timestamp precision to milliseconds (ISO-8601 format).	Improves forensics and settlement.

Recommended Actions

- Apply hotfix to inverters in serial ranges flagged in vendor change tickets (see Section 3).
- Monitor for increased TRIP-31 during firmware transition windows; adjust triage thresholds to avoid unnecessary escalations.

2.3 Rollback Conditions

Vendor recommended rollback triggers (apply immediately if observed):

- More than 10% of site inverters report additional downtime >= 5 minutes within 2 hours post-upgrade.
- Mass concurrent TRIP-44 events with *trip_correlated* flag true affecting >5 inverters at a site.
- Unrecoverable restart loops where inverter reboot attempts exceed 20 within 30 minutes.

Rollback steps: disable upgrade job in vendor portal, remotely stage v3.14.2 using vendor rollback utility, validate with acceptance tests (see Appendix: Validation Steps).

3. Vendor Change Tickets (2025-11-20 → 2025-12-08)

This section summarizes vendor change tickets and scope for the rollout and hotfix windows. These are synthetic but realistic records tied to the incident.

Ticket ID	Window	Scope (Plants)	Inverter Serial Ranges	Notes
VEND-CHG-2025-1120	2025-11-20 00:00 - 06:00 UTC	Plant A, Plant C	XG440-1000 → XG440-1299	Staged v3.14.2; pilot batch 50 inverters at Plant A.
VEND-CHG-2025-1121	2025-11-21 00:00 - 06:00 UTC	Plant B	XG440-1300 → XG440-1549	Full rollout for Plant B; monitoring for TRIP-31 increases.
VEND-HOTFIX-2025-1208	2025-12-08 02:00 - 04:00 UTC	Plant A, Plant B	XG440-1000 → XG440-1549	Applied v3.15.0 hotfix to address TRIP-44 misclassification.

Post-Change Monitoring Summary

Vendor recorded elevated TRIP-31 counts at Plant B in the 4 hours after the 2025-11-21 rollout; change ticket VEND-CHG-2025-1121 was updated to add extended monitoring. The 2025-12-08 hotfix addressed the majority of TRIP-44 false positives. See Section 4 for O&M incident timeline.

4. O&M Incident Report & Postmortem

Incident tied to the 2025-11-20 → 2025-11-21 rollout where firmware changes increased protective-trip signaling and led to curtailment windows and imbalance exposure. The following timeline and analysis are synthesized to include the required details for RAG use cases.

4.1 Executive Incident Timeline (Day-by-day)

Date (UTC)	Timeline
2025-11-20	Pilot upgrade at Plant A (50 inverters). No immediate anomalies detected.

2025-11-21	Full rollout at Plant B. From 01:15 to 03:00 UTC, Plant B recorded a spike in TRIP-31 alarms. By 03:30 UTC, several inverters entered repeated restart loops. Curtailment window declared 03:45–05:15 UTC to secure grid stability and reduce imbalance exposure.
2025-11-22	Reactive measures: manual firmware rollback requested. Vendor investigated and scheduled hotfix 2025-12-08.
2025-12-08	Hotfix v3.15.0 applied. TRIP-44 misclassification resolved; system returned to baseline by 07:00 UTC.

4.2 Curtailment Window Description

Curtailed generation: 03:45–05:15 UTC on 2025-11-21. Curtailment initiated to prevent cascading grid issues after identification of widespread TRIP-31 events indicating degraded grid ride-through performance. Curtailment type: dispatch curtailment via active disconnect and manual ramp-down through plant control.

4.3 Root Cause Analysis

1. Primary cause: The v3.14.2 change to debounce timing and standardized trip mapping altered how short transients were classified, resulting in some grid transients being interpreted as sustained ride-through failures (TRIP-31).
2. Secondary cause: High frequency of firmware upgrades across many inverters reduced the margin for transient handling during a simultaneous grid event, causing correlated trips (*trip_correlated* flag set) across multiple units.

4.4 Corrective Actions Implemented

- Applied hotfix v3.15.0 (VEND-HOTFIX-2025-1208) to refine TRIP-44 logic and add multi-channel correlation detection.
- Adjusted dispatch triage thresholds (see Section 5.2) to require higher counts per hour before automatic part replacement work orders are created.
- Updated automated ticketing to include *availability_minutes_lost* and *trip_correlated* fields for forensic and settlement processing.

4.5 Validation Steps (Post-Hotfix)

1. Run synthetic grid event simulation at Plant A with ramp disturbances; validate that TRIP-31 counts remain within baseline (<2 events per 24h per inverter).
2. Confirm *trip_correlated* flag behavior: trigger two inverters within 120s and ensure flag propagates.
3. Verify event_timestamp resolution millisecond precision in exported logs.

5. SCADA-to-CMMS Dispatch Runbook

This section defines the mapping rules from SCADA alarms and telemetry to CMMS corrective work orders, decision trees for triage, and sample automation payloads to the CMMS API. The SCADA

field names below must be used verbatim in automation systems:

- scada_alarm_code (string) — canonical alarm code (e.g., TRIP-31, TRIP-44)
- inverter_serial (string) — inverter serial identifier (e.g., XG440-1234)
- event_timestamp (ISO-8601 string with millisecond precision)
- availability_minutes_lost (integer) — minutes of unavailability attributed to the event
- trip_correlated (boolean) — true if >1 inverter at site tripped within correlation window

5.1 Alarm-to-Template Mapping

Mapping table below provides the CMMS template to be used and pre-filled fields for typical inverter protective-trip scenarios.

SCADA Alarm Code	Symptom	CMMS Template	Priority	Default Labor Code
TRIP-31	Grid ride-through protective trip; inverter offline, often auto-restores within 5-30m	Inverter_ProtectiveTrip_GridRideThrough	P2 (High)	LBR-INVG-01
TRIP-44	Internal hardware protection trip; may indicate over-temp or hardware faults	Inverter_ProtectiveTrip_Hardware	P1 (Critical) if trip_correlated=true or availability_minutes_lost >= 30	LBR-INVG-02
TRIP-31 (repeat)	Repeated TRIP-31 > threshold	Inverter_Repeated_Trip_Investigation	P2 → escalate to P1 if >3 occurrences/hr	LBR-INVG-03

Note: Parts are suggested—do not auto-issue parts without technician validation unless repeat count and availability loss thresholds are met (see Triage Rules).

5.2 Decision Trees & Triage Rules

Use the following decision tree for automated dispatch logic. The automation should evaluate SCADA events in near real-time (max 60s processing latency).

Decision Flow: Protective Trip Event

Step	Rule / Condition	Action
1	Receive SCADA alarm with scada_alarm_code (TRIP-31 or TRIP-44)	Log event; attach fields: inverter_serial, event_timestamp, availability_minutes_lost, trip_correlated
2	If scada_alarm_code == TRIP-44 AND trip_correlated == true	Create CMMS work order using template Inverter_ProtectiveTrip_Hardware with priority P1; send immediate escalation to on-call.
3	If scada_alarm_code == TRIP-31 AND availability_minutes_lost < 15 AND occurrences_per_hour <= 2	Create informational ticket (P3) and schedule remote diagnostics; do not dispatch unless condition escalates.
4	If scada_alarm_code == TRIP-31 AND occurrences_per_hour > 3 OR availability_minutes_lost >= 30	Create CMMS Inverter_Repeated_Trip_Investigation (P2) and assign onsite technician dispatch; include recommended parts if repeated failures >= 2 devices
5	If >5 inverters in site report protective-trip within 120s (trip_correlated true across devices)	Escalate to Site Manager and Grid Compliance team; consider immediate partial plant shutdown to limit imbalance exposure.

Triage Thresholds (Default)

- Occurrences per hour per inverter: Informational <= 2, Investigate > 3, Escalate >= 5
- Availability Minutes Lost: Informational < 15, Dispatch technician >= 30, Critical >= 120
- Concurrent alarms site-wide: Correlated window 120s; if correlated count >= 5 → treat as site event

Immediate Safety-Critical Actions

1. If P1 is triggered, notify on-call immediately and place phone call to Site Manager and Grid Compliance (see Appendix: Contacts). Do not wait for technician confirmation.
2. If site-level trip_correlated && grid operator requests disconnect, follow grid operator instructions and log curtailment window start/end times in CMMS ticket (fields: curtailment_start, curtailment_end).

5.3 CMMS Corrective Work Order Templates

Below are recommended template fields and default values for CMMS automated ticket creation. Field names should match CMMS API expectations.

Template	Field	Type	Default / Mapping
Inverter_ProtectiveTrip_GridRideThrough	title	string	"[TRIP-31] Inverter Protective Trip - {inverter_serial}"

Inverter_ProtectiveTrip_GridRideThrough	description	text	Include scada_alarm_code, event_timestamp, availability_minutes_lost, recent_restart_count, site_id
Inverter_ProtectiveTrip_GridRideThrough	priority	enum	P2 (High)
Inverter_ProtectiveTrip_Hardware	title	string	"[TRIP-44] Inverter Hardware Protection Trip - {inverter_serial}"
Inverter_ProtectiveTrip_Hardware	priority	enum	P1 if trip_correlated==true or availability_minutes_lost >= 30; otherwise P2
All Templates	custom_fields	map	scada_alarm_code, inverter_serial, event_timestamp, availability_minutes_lost, trip_correlated, site_id, vendor_firmware_version

Be explicit: always populate scada_alarm_code and inverter_serial to assist later RAG queries.

5.4 Sample Automation Payloads (CMMS API)

The following JSON payloads are sample requests to create CMMS work orders. Replace placeholders with live values.

Example A: TRIP-31 Informational Ticket

```
{
  "template": "Inverter_ProtectiveTrip_GridRideThrough",
  "title": "[TRIP-31] Inverter Protective Trip - XG440-1234",
  "description": "Automated ticket created from SCADA. scada_alarm_code:
TRIP-31. Observed auto-restore within 12m. Please run remote diagnostics before
dispatch.",
  "priority": "P3",
  "custom_fields": {
    "scada_alarm_code": "TRIP-31",
    "inverter_serial": "XG440-1234",
    "event_timestamp": "2025-11-21T03:17:02.123Z",
    "availability_minutes_lost": 12,
    "trip_correlated": false,
    "vendor_firmware_version": "v3.14.2",
    "site_id": "PLANT-B"
  },
  "assign_to_queue": "RemoteDiagnostics"
}
```

Example B: TRIP-44 Critical Work Order (Escalate)

```
{
  "template": "Inverter_ProtectiveTrip_Hardware",
  "title": "[TRIP-44] Inverter Hardware Protection Trip - XG440-1087",
```

```
  "description": "Automated P1 ticket. trip_correlated==true across >5 devices.
Immediate onsite dispatch and grid compliance notification required.",
  "priority": "P1",
  "custom_fields": {
    "scada_alarm_code": "TRIP-44",
    "inverter_serial": "XG440-1087",
    "event_timestamp": "2025-12-08T02:12:44.987Z",
    "availability_minutes_lost": 45,
    "trip_correlated": true,
    "vendor_firmware_version": "v3.15.0",
    "site_id": "PLANT-A"
  },
  "parts_to_hold": ["SKU-XG440-POWERBOARD"],
  "assign_to_queue": "O&M-Oncall"
}
```

API Integration Notes

- Use event_timestamp in ISO-8601 with milliseconds: YYYY-MM-DDThh:mm:ss.SSSZ
- Map scada_alarm_code verbatim to CMMS custom fields to ensure RAG retrieval accuracy.
- Include availability_minutes_lost to calculate potential settlement exposure and to prioritize field dispatch.

6. Troubleshooting & Field Procedures

6.1 On-Call Troubleshooting Checklist

1. Validate the SCADA alarm: confirm scada_alarm_code, inverter_serial, and event_timestamp are present.
2. Check recent firmware version on inverter (vendor_firmware_version). If running v3.14.2 and experiencing repeated TRIP-31 across site, consult vendor ticket history.
3. Verify trip_correlated flag across site-level telemetry.
4. Confirm availability_minutes_lost; if >= 30, prepare for onsite dispatch.
5. If restart attempts exceed 20 within 30 minutes, issue P1 and coordinate rollback with vendor.

6.2 Field Technician Procedure (Onsite)

Follow these steps when dispatched for a P2/P1 inverter protective trip investigation.

1. Pre-visit: Review CMMS ticket details and SCADA logs. Confirm parts list and safety gear.
2. Arrival: Safety brief with site personnel; record curtailment state and any grid operator instructions.
3. Diagnostics:

    - Check inverter local logs for scada_alarm_code, vendor_firmware_version, and controller reboot count.
    - Measure inverter temperature; if >85°C sustained for >30s, tag

TRIP-44 hardware fault.
    - Confirm DC/AC connections and inspect power board for visible damage if TRIP-44 confirmed.

4. Action:
    - If transient, re-enable inverter and monitor for 30 minutes before closing ticket as informational.
    - If hardware fault, replace module per SKU, update CMMS with parts used, downtime, and corrective action code.
5. Post-visit: Attach local logs (including event_timestamp, availability_minutes_lost) to CMMS ticket and set resolution status. Notify dispatch that ticket is updated.

6.3 Troubleshooting Flowchart (Textual)

Step A: Identify scada_alarm_code → If TRIP-44 → measure temp & check trip_correlated. If temp >85°C and sustained → replace hardware. Else examine logs for transient. Step B: If TRIP-31 → check occurrences_per_hour and availability_minutes_lost → follow triage thresholds in Section 5.2.

7. Settlement & Imbalance Charge Guidance (ERCOT)

This section explains how penalties are calculated in ERCOT for lost generation, and which data fields should be reconciled. It provides concrete guidance for post-event settlement reconciliation.

7.1 Key Concepts

- Imbalance settlements in ERCOT are based on the difference between scheduled exports and actual metered generation (metered energy).
- Availability_minutes_lost reported by inverters is an input for determining actual generation reduction contributions for each interval.
- Hourly and sub-hourly granularity matters — inverters report availability at 10s resolution; aggregate to settlement interval (e.g., 5-minute or 15-minute) as required by market settlement.

7.2 Data Fields to Reconcile

Field	Description	Usage in Reconciliation
event_timestamp	Exact time of trip	Map to settlement interval to attribute lost generation to correct settlement hour
inverter_serial	Which inverter experienced outage	Map to metered device mapping for plant
availability_minutes_lost	Minutes offline	Convert to equivalent MWh using inverter capacity and efficiency
scada_alarm_code	Cause of outage	Used for disputes and to justify exemption requests

7.3 Penalty Calculation Example (Simplified)

Assumptions:

- Inverter rated capacity: 500 kW
- Availability minutes lost: 30 minutes
- Settlement interval: 15 minutes

Energy lost = (500 kW) * (30/60 hours) = 250 kWh = 0.25 MWh. Multiply by the imbalance price in the affected settlement interval(s) to estimate exposure. Ensure that event_timestamp and availability_minutes_lost are accurately supplied to support settlement claims and potential exemption requests.

7.4 Documentation to Support Exemption or Dispute

1. Export SCADA logs with event_timestamp, scada_alarm_code, inverter_serial, availability_minutes_lost, and vendor_firmware_version.
2. Attach CMMS ticket with corrective actions and timestamps of technician arrival/repair.
3. Provide evidence of vendor rollbacks or hotfixes if event was caused by deployed firmware changes.

8. Appendices: Field Definitions, Error Codes, Contacts

8.1 Field Definitions (Canonical)

Field	Type	Definition / Allowed Values
scada_alarm_code	string	Canonical alarm code reported by inverter (e.g., TRIP-31, TRIP-44)
inverter_serial	string	Unique device serial, e.g., XG440-1234
event_timestamp	string (ISO-8601)	Timestamp with millisecond precision: 2025-11-21T03:17:02.123Z
availability_minutes_lost	integer	Number of minutes the inverter was not available as a result of the event
trip_correlated	boolean	True if correlation detected across multiple inverters within 120s

8.2 Error Codes & Remediation

Error Code	Symptom	Likely Root Cause	Resolution
E-TRIP-31	TRIP-31 reported; inverter offline; auto-restore	Grid transient or ride-through parameter sensitivity	Monitor; escalate if repeat events exceed threshold. Consider vendor debounce parameter adjustment.

E-TRIP-44	TRIP-44 hardware protection trip	Over-temperature or internal hardware fault	Inspect for overtemp; if sustained >85°C, swap power board. Use hardware diagnostic tools.
E-RST-LOOP	Frequent restart attempts	Firmware logic bug or power cycling	Place inverter in maintenance mode; escalate to vendor; consider rollback to previous firmware.

8.3 Change Ticket & Incident References

Reference tickets: VEND-CHG-2025-1120, VEND-CHG-2025-1121, VEND-HOTFIX-2025-1208. O&M incident dates: 2025-11-20 → 2025-12-08. Vendor firmware versions referenced: v3.14.2, v3.15.0.

8.4 Escalation & Contact Directory

Role	Name / Team	Contact (Phone / Email)	Escalation Criteria
Site Manager	SiteOps Lead	+1-555-0101 / siteops@plant.example.com	Any P1 event or trip_correlated affecting >=5 inverters
Grid Compliance	GridTeam	+1-555-0102 / gridcompliance@example.com	Any curtailment window or interaction with ERCOT
Vendor Support	XG Vendor Oncall	+1-800-555-2211 / support@xgvendor.example.com	Firmware issues, rollback requests, and hotfixes
Dispatch Desk	Dispatch	+1-555-0103 / dispatch@example.com	All automated CMMS escalations and technician scheduling

8.5 Validation Checklist (Post-Repair & Post-Upgrade)

1. Confirm inverter reports vendor_firmware_version recorded in CMMS and SCADA.
2. Validate that event_timestamp is present and millisecond-precise for each recorded trip.
3. Reconcile availability_minutes_lost against plant metering for the settlement interval.
4. Run acceptance test: simulate a transient and verify no false TRIP-44 escalation.

Distribution: Internal use. Contains vendor incident synthesis for RAG indexing. When integrating into automation, ensure secure credential storage for CMMS API and vendor portal interactions.